Master PRD v2.0 — Mahalaxmi Kirana: AI Order & Inventory Platform

Team NUCLEUS · Client: Mahalaxmi Kirana Store, Yesgaon, Tal. Kopargaon · Status: Draft for review · Supersedes the earlier v1.0 PRD

1. Product Overview

Vision. A Marathi-first ordering and inventory platform that a real village kirana store runs its daily business on. Customers browse and order on the website, or type or speak a grocery list that AI converts into a cart. The owner controls products, stock, orders and credit (udhar) from a phone-friendly dashboard, and sees every important event in an in-website message box.

Goals

1. Real, live use at the shop: every number on screen (price, stock, order status) is true.
2. Zero overselling: stock is numeric and reduced atomically at order time.
3. Marathi first, with Hindi and English available through a language switcher.
4. AI ordering that is helpful but never invents products or prices, and never saves an order without customer confirmation.
5. Owner can change anything about the catalog in real time, with no developer involved.
6. Runs on free-tier hosting with graceful behavior when the free server sleeps.

Non-goals (v2.0): online payment gateway, delivery-partner assignment or live tracking, loyalty points and coupons, multi-branch inventory, WhatsApp Business API (automatic WhatsApp messages), customer ratings system.

Success metrics (first 90 days live)

- Oversell incidents: 0
- Share of shop orders arriving through the platform: tracked weekly
- AI order box: share of parsed items accepted without edit (target above 80%)
- Time from order placed to owner confirming: median under 10 minutes in shop hours
- Low-stock items noticed by the owner before hitting zero: tracked via inbox read time

2. Users and Roles

Role	Description	Access
Customer	Residents of Yesgaon and Kopargaon area, many on basic Android phones, Marathi-speaking	Browse, AI order box, cart and checkout, order history, message box, udhar balance (if approved), language switch
Owner	Mr. Kiran Gaikwad	Everything, including staff management and credit approvals
Staff	Added by the owner	Same screens as the owner, except adding or removing staff and changing the owner's login. Every staff action is logged with the staff member's name

3. Current State (from the code audit, Oct 15 2026)

Already built and kept: bilingual storefront, product catalog with categories, cart, 3-step checkout (address, delivery slot, payment by UPI QR or COD), order lifecycle (new, confirmed, packed, out for delivery, delivered), customer cancellation, reorder, owner dashboard, packing list, settings (minimum order, delivery fee, free-delivery threshold, slots, serviceable pincodes, UPI details), Cloudinary uploads, bcrypt and JWT auth, rate limiting, 24 integration tests.

Gaps this PRD closes: inventory is only an in-stock flag; no stock deduction or logs; no low-stock alerts; no AI; WhatsApp is only a link; no best sellers; reports are today-only; no udhar; no notification inboxes; server can crash on a database blip; no backups or deployment setup.

4. Decisions Log

#	Decision
D1	Stack stays Node/Express + MongoDB. The college report is updated to match

#	Decision
D2	No WhatsApp API. Customers get no automatic WhatsApp messages
D3	After checkout the order is saved first; a button then opens the owner's WhatsApp chat with the bill pre-filled
D4	Important messages go to in-website message boxes (owner and customer), plus optional web push
D5	Owner approves which customers get udhar; reminders use pre-filled WhatsApp messages
D6	Customers cannot edit or add items after placing an order
D7	Out-of-stock substitutes are offered
D8	Languages: Marathi (default), Hindi, English, with a language switcher and strongest Marathi quality
D9	AI: typed input first; Marathi voice input is a planned feature; ambiguity is resolved with option buttons
D10	Free-tier hosting for now; Gemini API key supplied by the team
D11	Testimonials are labeled sample content and managed from admin
D12	Catalog is fully owner-managed in real time (no fixed product list)

5. Feature Requirements

Priority: **P0** must have for go-live, **P1** important, **P2** later. Release column refers to Section 11.

5.1 Live Catalog Management (CAT)

ID	Requirement	Pri	Rel
CAT-01	Owner and staff can add, edit, hide and delete products and categories; changes appear to customers without any redeploy	P0	R1
CAT-02	Product fields: names in Marathi and English (required), Hindi (optional), aliases (for example साखर, शक्कर, sugar), category, unit type (piece, kg, g, litre, ml, pack), pack size, price, MRP, image, offer percent, active flag, reorder level	P0	R1
CAT-03	Customers see current price and availability on every page load; the cart is re-validated against live price and stock at checkout; any change is shown clearly before the customer pays	P0	R1
CAT-04	Orders store a price snapshot, so later price edits never alter past orders	P0	Done
CAT-05	Bulk import and export of products by Excel or CSV, with a preview and error report before saving	P1	R1
CAT-06	Offers and Mega Deals are driven by owner-set offer values and optional end time; no code changes	P1	R2
CAT-07	Best Sellers computed from the last 30 days of orders and shown on the home page	P1	R2
CAT-08	Search tolerates spelling variants and mixed scripts (Marathi, Devanagari Hindi, English letters) using aliases and fuzzy matching	P0	R3
CAT-09	Category and product ordering is controlled by the owner (drag or number)	P2	R4

5.2 Inventory (INV)

ID	Requirement	Pri	Rel
INV-01	Numeric stock per product, with decimals allowed for loose items (for example 2.5 kg)	P0	R1
INV-02	Order creation reduces stock with a conditional atomic update; if any item is short, nothing is reduced and the customer sees what is available	P0	R1
INV-03	Cancellation restores stock exactly once, even on double-clicks	P0	R1
INV-04	Stock log for every change: in, out, adjustment, order, cancel, with balance after, reason note and who did it	P0	R1
INV-05	Low-stock list and dashboard badge when stock is at or below the product's reorder level	P0	R1
INV-06	Fast stock-in screen designed for phone use (search product, enter quantity, save)	P0	R1
INV-07	Customers see Only N left for low stock and cannot add zero-stock items	P0	R1
INV-08	Optional expiry date per batch with near-expiry list	P2	R5
INV-09	Owner can mark an item temporarily unavailable regardless of stock	P1	R1

5.3 Ordering and Checkout (ORD)

ID	Requirement	Pri	Rel
ORD-01	Keep current checkout: address, delivery slot, payment (UPI QR, COD, Udhar for approved customers)	P0	Done
ORD-02	Serviceable pincodes, minimum order and delivery fee rules come from owner settings	P0	Done

ID	Requirement	Pri	Rel
ORD-03	Customers can cancel only while the order is new or confirmed; stock is restored	P0	Done
ORD-04	One-tap reorder from history, checked against current price and stock	P0	Done
ORD-05	At cart time, if an item is short or out, the customer is offered substitutes (same category or owner-defined alternatives) with Replace or Remove buttons	P1	R3
ORD-06	After an order, only the owner or staff may change items. If they swap an item, the customer gets a message-box notice with Accept or Reject buttons for the substitute	P1	R5
ORD-07	Orders carry a note field and an optional map pin or landmark for village addresses	P1	R2
ORD-08	Each order records channel: Website, AI order box	P1	R3

5.4 AI Order Box (AI)

Experience. On the home page and cart page: a box labeled आपली यादी टाईप करा / बोला (Type or speak your list). Example input: 2 kilo sakhar, 1 litre goda tel, 1 bath soap. The system returns a cart for review. It never places the order itself.

ID	Requirement	Pri	Rel
AI-01	Customer types a list in Marathi, Hindi, Hinglish or English; the backend sends it with a compact product list to the Gemini model and receives structured JSON (item text, matched product IDs, quantity, unit, confidence)	P0	R3
AI-02	Server validates every returned product ID against the database, recomputes prices and checks stock; the model's output is never trusted for price or availability	P0	R3

ID	Requirement	Pri	Rel
AI-03	Ambiguous items (for example two kinds of oil) are shown as option buttons; unmatched items are listed as not found with a search shortcut	P0	R3
AI-04	Parsed items fill the cart for review; the customer must confirm and complete checkout	P0	R3
AI-05	Fallback: if the AI service fails or its quota is exhausted, fuzzy matching on names and aliases is used and the customer is told the AI is busy	P0	R3
AI-06	Per-customer daily limit and per-request size limit; the API key lives only on the server	P0	R3
AI-07	Marathi voice: record up to 30 seconds, backend transcribes (Marathi, Hindi, English), the text is shown for correction, then parsed as above; the browser's own speech recognition is the fallback; audio is discarded after transcription	P1	R5
AI-08	Owner dashboard shows which phrases the AI could not match, so the owner can add aliases and improve accuracy	P1	R4
AI-09	Reply language follows the customer's chosen language	P1	R3

Model names, audio support and free-tier limits are checked against Google's current documentation at build time.

5.5 WhatsApp-Assisted Actions (WA)

All WhatsApp use is through click-to-chat links (wa.me) with pre-filled text. There is no backend WhatsApp integration.

ID	Requirement	Pri	Rel
WA-01	After an order is saved, the confirmation page shows Send bill to owner on WhatsApp; the message holds order number, items, quantities, total, address, slot, payment mode and a map link	P0	R2

ID	Requirement	Pri	Rel
WA-02	The order exists in the database and dashboard whether or not the customer taps the button	P0	R2
WA-03	Owner can open a pre-filled WhatsApp message to a customer from an order (for example a Marathi status line)	P1	R2
WA-04	Owner can share today's summary and the low-stock list through pre-filled WhatsApp text	P1	R4
WA-05	Udhar payment reminder as a pre-filled WhatsApp message in the customer's language	P1	R4
WA-06	A floating contact button on the storefront opens a chat with the shop	P1	Done

5.6 Message Boxes and Notifications (MSG)

Two in-website inboxes replace automatic WhatsApp messages.

Owner and staff inbox events: new order, order cancelled by customer, low stock or out of stock, new udhar request, customer accepted or rejected a substitute, AI phrases that failed repeatedly, daily summary.

Customer inbox events: order placed, confirmed, packed, out for delivery, delivered, cancelled by shop (with reason), substitute proposed, udhar approved or declined, udhar payment recorded, owner announcements (rate-limited).

ID	Requirement	Pri	Rel
MSG-01	Notification collection with type, title and body in the recipient's language, link to the related order or product, read state and timestamp	P0	R2
MSG-02	Bell icon with unread count and an inbox page on both customer site and admin	P0	R2
MSG-03	Owner inbox keeps the existing chime and 30-second refresh	P0	Done
MSG-04	De-duplication: one low-stock message per item until it is restocked	P0	R2

ID	Requirement	Pri	Rel
MSG-05	Optional web push (opt-in) so alerts reach the phone when the site is closed; on iPhone this needs the site added to the home screen	P1	R2
MSG-06	Actionable messages (Accept or Reject substitute) work from the inbox	P1	R5
MSG-07	Messages older than 90 days are archived automatically	P2	R6

5.7 Udhar Credit Ledger (UDH)

ID	Requirement	Pri	Rel
UDH-01	Customer can request credit; only owner-approved customers can choose Udhar at checkout	P0	R4
UDH-02	Owner sets a credit limit per customer; checkout blocks credit orders beyond the limit	P0	R4
UDH-03	Ledger entries: credit order, payment (cash or UPI), adjustment, each with date, note and who recorded it; running balance per customer	P0	R4
UDH-04	Customer sees only their own balance and entries	P0	R4
UDH-05	Owner dashboard lists customers by outstanding amount and age of dues	P1	R4
UDH-06	Owner can pause or revoke credit for a customer	P1	R4

5.8 Owner Dashboard, Staff and Reports (ADM)

ID	Requirement	Pri	Rel
ADM-01	Owner can add and remove staff accounts; staff see the same screens as the owner except staff management and owner credentials	P0	R4

ID	Requirement	Pri	Rel
ADM-02	Audit log of staff and owner actions (price changes, stock edits, order status, credit changes)	P0	R4
ADM-03	Reports with date range: orders, sales, delivery fees, payment split (cash, UPI, udhar), top and slow products, stock value	P1	R4
ADM-04	CSV export for orders, stock log and udhar ledger	P1	R4
ADM-05	Dashboard home: today's orders, pending count, low-stock count, unread messages, udhar outstanding	P0	R2
ADM-06	Manage sample testimonials: add, edit, hide, with a visible Sample label on the site	P1	R2
ADM-07	Packing list and order queue remain, with filters by status and slot	P0	Done

5.9 Language and Accessibility (LNG)

ID	Requirement	Pri	Rel
LNG-01	Marathi default; switcher for Marathi, Hindi, English; choice saved per customer and device	P0	R0
LNG-02	All interface text lives in translation files, not in the page code	P0	R0
LNG-03	Marathi text is reviewed by a native speaker and uses natural everyday wording rather than literal translation	P0	R6
LNG-04	Large touch targets, readable Devanagari font sizes, high contrast, usable on low-end Android	P0	R2
LNG-05	Admin can use the dashboard in Marathi or English	P1	R4

6. Data Model Changes

Collection	Change
Product	Add stockQuantity, reorderLevel, unitType, aliases[], nameHi (optional), unavailableFlag. inStock becomes derived
StockLog (new)	product, type, quantity, balanceAfter, note, actor, order, createdAt
Notification (new)	recipient (user or owner group), type, title, body, link, readAt, dedupeKey, createdAt
PushSubscription (new)	user, endpoint and keys, createdAt
UdharAccount (new)	customer, approved, creditLimit, balance, status
UdharEntry (new)	account, type, amount, note, order, actor, createdAt
User	Add role staff, preferred language
AuditLog (new)	actor, action, target, before and after summary, createdAt
Order	Add channel, note, location pin, payment mode includes udhar, substitution records
AiLog (new)	customer, input text, matched items, unmatched phrases, accepted or edited flags (no audio stored)
Testimonial (new)	text, name, area, visible, sample flag

7. Free-Tier Hosting and Operations

Planned stack: static frontend on Vercel, Node server on Render free tier, MongoDB Atlas free cluster, Cloudinary free tier for images, Gemini free tier for AI. Exact limits of each free plan change over time and must be checked before go-live.

Concern	Approach
Server sleeping when idle	Free web services typically sleep after inactivity and take time to wake. Mitigate with a scheduled pinger during shop hours, a friendly loading state on first request, request retry with backoff, and a cached app shell and last-seen catalog for browsing
No automatic backups on free database	Scheduled export of key collections to private storage, plus a manual Download backup button for the owner
Free quotas (AI, images, database size)	Server-side rate limits, compact AI prompts, image compression before upload, usage counters on the admin settings page
Secrets	All keys in server environment variables only; none in the frontend or repository
Upgrade path	Code kept portable so moving to a paid always-on server later needs only configuration changes

8. Non-Functional Requirements

- **Security:** hashed passwords, JWT with expiry, role checks on every admin route, input validation, NoSQL injection protection, locked-down CORS to the real domains, rate limits (including AI and voice endpoints), HTTPS everywhere, audit log.
- **Privacy:** privacy policy page, customer can request account and data deletion, minimal data kept, audio never stored, phone numbers and addresses visible only to owner, staff and the customer. Review current Indian data-protection requirements before go-live; this document is not legal advice.
- **Reliability:** process-level error handlers so a database blip never crashes the server; atomic stock updates; idempotent order creation to prevent duplicates from double taps; clear error messages in the user's language.
- **Performance:** catalog loads quickly on mobile data; images compressed and lazy-loaded; search responds instantly on the client; first load on a cold server shows a friendly state within 2 seconds.

- **Maintainability:** automated tests for stock, orders, udhar, roles and AI validation; README with setup, environment variables and backup restore steps.
- **Compatibility:** modern Android Chrome first, then iOS Safari and desktop.

9. Key User Flows

A. Customer orders by typing a list. Opens site (Marathi) → types the list → AI shows cart with option buttons for ambiguous items → customer fixes and confirms → checkout (address, slot, payment) → server checks stock and creates order → confirmation page with Send bill to owner on WhatsApp → customer inbox shows order placed.

B. Owner handles an order. Chime and push alert → inbox and dashboard show new order → owner confirms → customer inbox shows confirmed → packing list → packed → out for delivery → delivered (and udhar entry or payment recorded if relevant).

C. Owner restocks. Opens stock-in → searches product (any script) → enters quantity → stock log written → low-stock flag clears.

D. Udhar customer. Requests credit → owner approves with limit → customer chooses Udhar at checkout → ledger entry created → owner records payments later → owner sends a pre-filled reminder if dues are old.

10. Risks and Mitigations

Risk	Mitigation
Owner misses alerts when the site is closed	Optional web push, chime, dashboard badge, and a daily summary message
Free server sleeps and slows the first order of the day	Pinger, loading state, retry, cached shell
AI misreads a Marathi item	Server-side validation, option buttons, customer confirmation, fuzzy fallback, owner-managed aliases
AI free quota runs out	Per-user limits, fallback matcher, usage counter
Wrong stock because of manual counting errors	Adjustment with notes, audit log, periodic stock-check screen

Risk	Mitigation
Staff misuse of owner-level access	Audit log with names, owner can disable any staff account instantly
Report and code mismatch at college evaluation	Report updated to Node/Express + MongoDB and WhatsApp-assisted scope (Section 12)
Customer data exposure	Role checks, locked CORS, minimal data, deletion flow

11. Release Plan

No fixed deadline: each release is finished, tested and used at the shop before the next begins.

Release	Scope
R0 Foundation	Crash handlers, config fixes for phone access, translation files and language switcher scaffold, audit log base
R1 Live inventory and catalog	Numeric stock, atomic deduction, stock logs, low-stock list, stock-in screen, aliases and unit types, bulk import, live price and stock re-validation
R2 Messaging and trust	Owner and customer message boxes, web push, WhatsApp bill-to-owner flow, order notes and map pin, best sellers, offers control, testimonial management, dashboard home
R3 AI ordering	Typed AI order box, fuzzy search, option buttons, fallback, limits, cart substitutes
R4 Owner power tools	Udhar ledger, staff accounts, audit log screens, reports with date range, CSV export, shareable summaries and reminders, AI miss list
R5 Voice and polish	Marathi voice order, full Hindi coverage, owner-proposed substitutions, expiry tracking, PWA offline browsing

Release	Scope
R6 Go-live hardening	Backups, tests, monitoring, native-speaker Marathi review, privacy pages, README, go-live checklist, report update

12. Academic Report Alignment

Report item	Update needed
Architecture (Java, Spring Boot, relational DB)	Change to Node/Express, MongoDB, vanilla JS frontend
ER and class diagrams	Redraw for new collections in Section 6
FR-07 WhatsApp enquiry	Reword to WhatsApp-assisted ordering: bill sent to owner through pre-filled chat
FR-08 AI message understanding	Keep, now implemented through the AI order box
FR-11 Confirmation	Web confirmation plus message boxes
FR-17 Low-stock alerts	Dashboard badge and message-box alert
Annexure testimonials	Mark as sample content
Out of scope	Add WhatsApp Business API, online payment, delivery tracking

13. Open Items

1. Final custom domain, if any.
2. Number of real products and the source of the first catalog import.
3. Wording of the privacy policy and cancellation policy, to be agreed with the owner.
4. Native-speaker reviewer for Marathi text.
5. Confirmation of current free-tier limits (Render, Atlas, Cloudinary, Gemini) at build time.